

Meeting with Dr. Songa, 21 September 2026

One card per slide. Read the script in your own words; the right column is the numbers to say out loud. Likely questions and a plain-words sheet follow.

01 Title 1 min

02 The question 1 min

03 Plan vs reality 1.5 min

04 Tools 1 min

05 Process 1 min

06 The articles 1 min

07 Groups differ 1 min

08 Which group 1 min

09 Repeatable? 1 min

10 Honest check 1.5 min

11 Real brains 1 min

12 The minimum push 1 min

13 Children's content 1 min

14 Limits 0.5 min

15 Papers and requests 1.5 min

Total about 16 minutes

Title

SAY

Good morning, Dr. Songa, and thank you for the time. I'll take you through the dissertation as it stands after your review: what I set out to do, the tools and steps, what we found, where the proposal turned out wrong, what the work cannot claim, and the three papers. About fifteen minutes, and I'll end with what I need from you.

NUMBERS TO SAY

- About 15 minutes in total
- End with the four requests

The question

SAY

Here is the idea in one picture. Physicists model a group forming an opinion the same way they model tiny magnets lining up. Each arrow is a person, and people tend to copy those around them. Media is an outside push on everyone at once. In every study I read, the size of that push is simply chosen by the researcher. My question was: can we measure it from the content itself?

NUMBERS TO SAY

- h = the push from media
- J = how much people copy each other

Plan vs reality

SAY

Before the results, I want to be upfront about what changed from the proposal. Five assumptions turned out differently once we tested the tools. The biggest: we planned to measure the amygdala, the brain's alarm centre, but the model only predicts the brain's outer surface. So we used emotion-linked and reasoning-linked areas on the surface instead. All of this is in the amendment you approved.

NUMBERS TO SAY

- 5 changes, all in the approved amendment
- Ratio broke for 69 of 400 articles
- Two physics coefficients corrected

Tools

SAY

These are the tools. The main one is TRIBE v2, published by Meta's research lab. It learned from brain scans of adult volunteers watching TV and films, and it predicts how a typical brain would respond to new content. Because it was built on people listening, we read each article aloud first. Nobody was scanned in this project. Every brain value you'll see is a prediction.

NUMBERS TO SAY

- TRIBE v2: Meta, 2025
- About 70 seconds per article on a P100
- No person was scanned

Process

SAY

This is the whole process for one article. It's turned into speech, the timing of each word is marked, and the model predicts activity at 20,484 points across the brain's surface. We average 1,030 points in areas linked to emotion and 851 in areas linked to careful reasoning. The score is emotion minus reasoning. Above zero, emotion leads. Below zero, reasoning leads.

NUMBERS TO SAY

- 20,484 points on the brain surface
- 1,030 emotion, 851 reasoning
- Score = emotion minus reasoning

The articles

SAY

We scored 400 articles from four public collections, 100 each: fear-driven fake news, outrage-style partisan news, clickbait, and neutral writing. We matched their lengths so no group wins by being longer, and we worked out before scanning that 400 was enough to see a small effect. Then we ran every article twice.

NUMBERS TO SAY

- 4 groups x 100 articles
- About 164 words each
- Smallest detectable effect 0.027
- Every article scanned twice

Groups differ

SAY

First result: the four groups really do differ. The number is eta squared, 0.107. In plain terms, about 11 percent of the differences between articles line up with their group. The rest is article-to-article variation, because the groups overlap a lot. The chance of this being luck is about one in a billion, and the second run gave 0.089, so it held.

NUMBERS TO SAY

- 0.107 = about 11% explained by group
- p = about 1 in a billion
- Second run 0.089

Which group

SAY

Which group drives it? Fear. Compared with neutral articles, fear-driven ones differ by 0.94, which counts as large. Clickbait is small. Outrage barely moves, which surprised us, because the proposal expected it to be strongest. Looking closer, outrage raises both the emotion and the reasoning areas by about the same amount, so the gap between them stays flat.

NUMBERS TO SAY

- Fear 0.94 (large)
- Clickbait 0.32 (small)
- Outrage 0.03: both areas rise, 0.51 and 0.49

Repeatable?

SAY

Can we trust the score? We ran all 400 articles twice. Agreement between runs is 0.87, where 1 means identical, so it's high. It isn't perfect because the speech voice and the computer's arithmetic vary slightly. 51 articles flipped sign. Noise alone predicts about 55, and those articles all sit near zero. So we only make claims about groups, never one article.

NUMBERS TO SAY

- Agreement 0.87 (1 = identical)
- 51 flipped, about 55 expected (44 to 67)

Honest check

SAY

Now the most important honest check. Simple word counting sorted manipulative from neutral articles far better than our score: 0.98 against 0.63, where 0.5 is a coin toss. The reason is that each group came from a different source, and the words give the source away. So we can't say the differences come from writing style alone. Our score is a measurement for the physics, not a manipulation detector.

NUMBERS TO SAY

- Word counting 0.98, ours 0.63, sentiment 0.54
- 0.5 = coin toss
- Source guessed 66% of the time vs 25% by chance

Real brains

SAY

Does the brain model match real brains at all? A company audit claimed it was the opposite of real brain data, so we tested it on public scans of people watching the show Friends. Real people's brains agree with each other at 0.152, the best any model can reach. On a two-minute clip the model scored 0.028, against a best possible of 0.096 on that clip. Weak, but positive. We also found and fixed three hidden bugs. This is Paper 3.

NUMBERS TO SAY

- Best possible 0.152 (full episode)
- Model 0.028 vs 0.096 on the 2-minute clip
- $p = 0.048$
- 3 bugs found and fixed

The minimum push

SAY

Here is the physics result. Media's push equals our score times a strength called alpha. We couldn't measure alpha from the data, so we don't quote one. Instead we asked: how big must alpha be for media alone to flip a majority? With our scores spread over 0.124, the answer is at least 4.29 when people copy each other strongly. Any future claim that content like this flips opinion has to clear that bar.

NUMBERS TO SAY

- Spread of scores 0.124 (-0.070 to +0.054)
- Alpha at least 4.29 at copying strength 2
- No value of alpha is claimed

Children's content

SAY

At my first presentation I was asked whether this could analyse any content, even what children watch. Technically, yes: the pipeline takes video and audio, so it could score a cartoon or an advert. But the model learned from adult brains, and children's brains are still developing, so it can't say how a child reacts. We also only tested news text. For children, we'd need child brain data and ethics approval first.

NUMBERS TO SAY

- Can: video and audio, groups of content
- Cannot: adult-trained model, 1 in 8 items flip
- Needs: child data and ethics approval

Limits

SAY

To be clear about the limits: every brain value is a prediction, not a scan. Group and source are mixed together in this dataset. The model can't see deep areas like the amygdala. We couldn't measure alpha. And the check against real brains used only two minutes of video. Each is stated in the thesis, and each points to the next piece of work.

NUMBERS TO SAY

- 5 limits, each stated in the thesis

Papers and requests

SAY

Three papers come out of this. Paper 1 is pure physics: the minimum-push rule, no brain data needed. Paper 2 is the measuring tool and the 400 articles, including the word-counting comparison. Paper 3 is the check against real brains. The dissertation comes first. I'd like any corrections you need, your signature on the declaration page, and clearance for the library by Thursday. And I'd like to ask about co-authorship on the papers.

NUMBERS TO SAY

- Bound copy to the library by Thu 24 Sep
- Graduation clearance by Mon 28 Sep
- Ask about co-authorship

Many of these grow out of the first presentation's question about children's content

Q1 Can this analyse any content, even what children watch?

A Technically it can score video and audio, not only text. But the model learned from adult brains, so it can't tell us how a child reacts, and we only tested news text. It would need child brain data and ethics approval first.

Q2 Could a parent or a regulator use it to rate one show?

A Not yet. About 1 in 8 items flip between runs, so a single score isn't reliable. It's only trustworthy for comparing groups of content, and not for children until it has been tested on them.

Many of these grow out of the first presentation's question about children's content

Q3 Was anyone scanned? Is it reading people's minds?

A No. Nobody was scanned. The scores are predictions from a published model of a typical adult brain. The tool rates content, not people, and knows nothing about any individual.

Q4 Could someone use it to make content more manipulative?

A That's a real risk with any measurement of content. That's why it's framed as a research measurement, reports only group results, and the model's licence allows non-commercial research only.

Many of these grow out of the first presentation's question about children's content

Q5 If word counting does better, why not use that?

A Word counting answers a different question: it mostly learns which source an article came from. Our score is meant to be the push in the opinion model. The comparison is there to be honest about what the score is not.

Q6 Why read the articles aloud?

A The model was trained on people watching and listening, so it expects speech with word timings. Reading aloud puts the article in the form the model understands.

Many of these grow out of the first presentation's question about children's content

Q7 Why no amygdala, when the proposal named it?

A The released model only predicts the brain's outer surface. The amygdala sits deep inside, so we can't measure it and make no claim about it. We used surface areas linked to emotion instead.

Q8 What would it take to work for children's media?

A Brain scans from children collected with ethics approval and consent, a test set of children's videos, content where source and type are mixed, and a longer check against real brains.

Many of these grow out of the first presentation's question about children's content

Q9 Why is this a physics project?

A The opinion model is statistical physics, the same maths as magnets lining up. The contribution is measuring the push in that model and working out the minimum strength it needs. The brain model is the measuring tool.

Q10 What does 'alpha at least 4.29' mean in practice?

A It's a bar to clear. If someone claims content like this flips a strongly connected group's opinion, their alpha must be at least 4.29, or the model says it can't happen. It does not say media does flip opinion.

Many of these grow out of the first presentation's question about children's content

Q11 Why only 400 articles?

A We calculated before scanning that 400 was enough to see an effect as small as 2.7%. Each article takes about a minute of GPU time, and we ran all of them twice.

Q12 Is it okay to use Meta's model?

A Yes, for research. It's released under a non-commercial licence, it's cited throughout, and this project makes no money from it.

If a symbol comes up, say it like this

η^2 , eta squared Share of the differences explained by group. 0.107 is about 11%.

p-value Chance a result this strong appears by luck. 1×10^{-9} is about one in a billion.

d, Cohen's d Size of a difference in units of normal spread. 0.2 small, 0.5 medium, 0.8 large.

ICC Agreement between two runs. 1 identical, 0 none. Ours 0.87.

AUC How well a score sorts two kinds of article. 0.5 coin toss, 1 perfect.

r, correlation How closely two things rise and fall together. 0 none, 1 perfect.

Noise ceiling Best score any model can reach, given how much real brains differ. Ours 0.152.

α , alpha Strength of media's push per unit of our score. Not measured.

β J How strongly people copy each other. Above 1, a group settles on a majority by itself.

ΔX Spread of our scores, lowest to highest: 0.124.

h with small c The critical push: what it takes to flip a majority.